

Shape A exterior arrangement

Slimmer grip with a taller top sampling pod. Single rear cover; screen-only front. Viewer-left power button and bottom USB-C.

ENGINEERING CONCEPT REVIEW

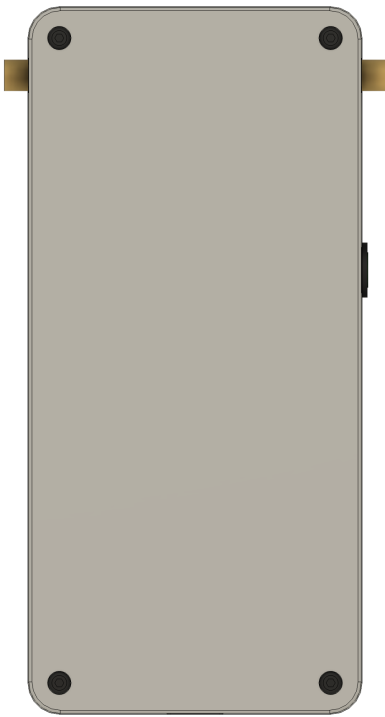
NOT TO SCALE / PRINT RELEASE PENDING

01 / FRONT



Measured 4.3-inch factory-cased display; top sampling pod above.

02 / REAR



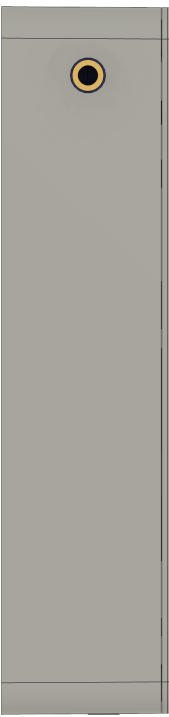
One removable cover with four separate recessed rear screws.

03 / LEFT SIDE



Power button on viewer-left side; upper gas fitting.

04 / RIGHT SIDE



Opposed upper gas fitting. USB-C is on the bottom face.

DIMENSION REFERENCES / MILLIMETRES

REFERENCE	ENVELOPE / SIZE	BASIS AND LIMIT
Housing	180 H x 85 W x 43 D	Audited total: 180 H x 97 W x 43 D, including projections.
Factory-cased display	116.8 H x 69.3 W x 13.7 D	Owner measurement with factory back removed; local mounting details unmeasured.
Occupied FMA holder	80.4 H x 42 W x 20.35 D	Owner measurement including two cells; retained RCY/BEC plug is provisional geometry.
Future PCB strip	30 W x 99 H, before notches	Datum-derived at 85 mm case width; footprint study only, not a routed board.

Shape tradeoff: 23.2% thinner than A2; 25.3% more body bounding-box volume. The measured factory display casing occupies 52.9% of the front rectangle; this is not active screen area.

Sources: owner measurements; [Guition specification](#); [GCT USB4720 Revision B drawing](#). Actual viewport views are independently resized; do not measure these sheets.

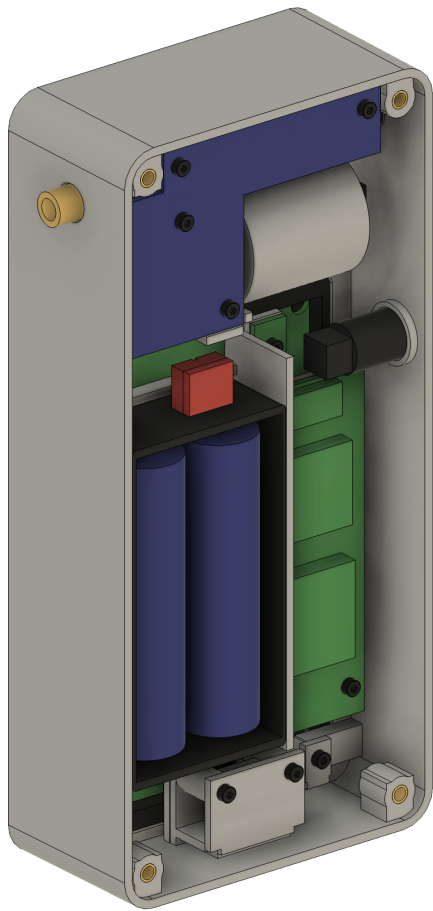
Separate battery, PCB and sample path

Battery and PCB sit side by side behind the display. The top manifold uses a continuous candidate clearance route; gas performance remains unqualified.

ENGINEERING CONCEPT REVIEW

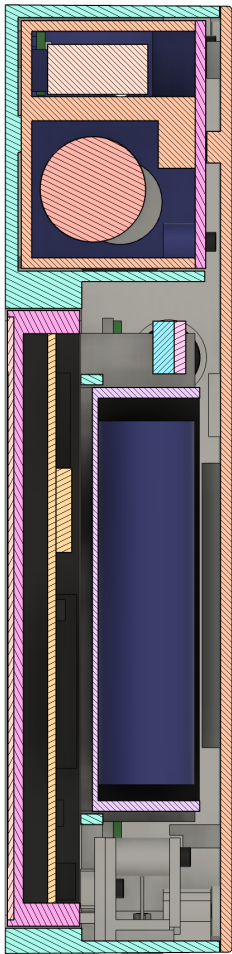
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05 / REAR OPEN



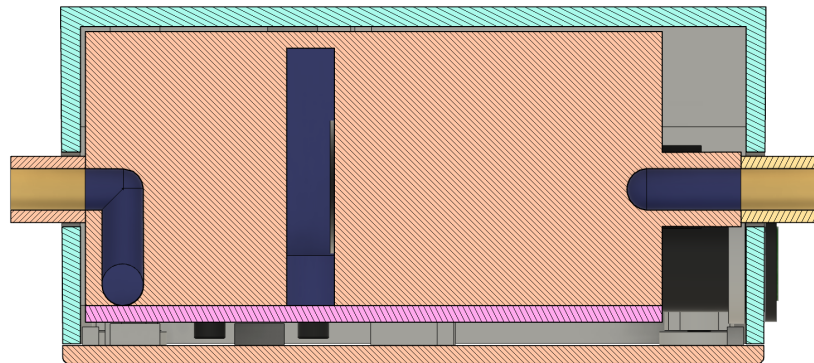
Cover hidden. Pack, side PCB strip and top cartridge remain visible.

06 / SECTION A-A



Battery/screen cut. Screen left; rear cover right.

07 / SECTION B-B



Top-manifold cut at GasY. Front at image top; rear at bottom.

READING THE MODEL

5 MM CONTINUOUS GAS PROBE / PASS / 0 INTERSECTIONS

+X points toward the viewer's left; +Y up; +Z rearward. Rear views reverse visual left/right. The complete display leaves forward after its rear retainers are released. Keep the protected cells in the measured holder.

Gas evidence: gas-clearance-paths.json records cylinders, corner spheres and selected channel dimensions. Flow rating, sample renewal, sealing and sensor response require physical validation. Nominal part allowances, the AO2 seal and atmospheric sampling remain unqualified.

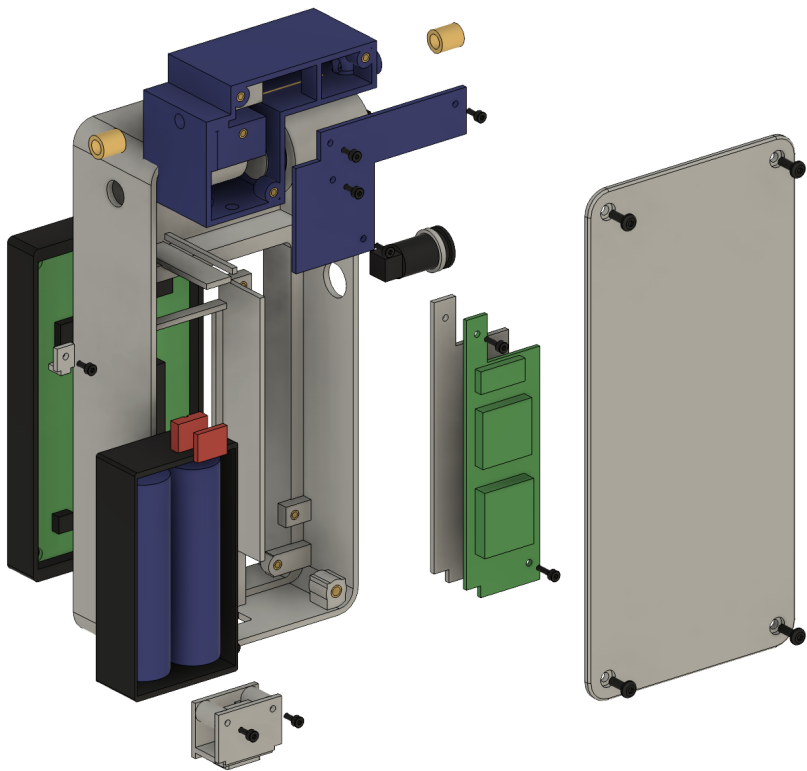
Exploded relationships and recorded checks

Separate occurrences for individual screws. The exploded arrangement explains ownership and access; it is not a removal trajectory.

ENGINEERING CONCEPT REVIEW

NOT TO SCALE / PRINT RELEASE PENDING

08 / EXPLODED ASSEMBLY



Cover, display, battery, carrier, PCB, USB cartridge and top sample assembly shown separately.

CURRENT REVISION04 EVIDENCE

Native CAD / PASS
102 bodies, 66 occurrences; 0 overlaps. 8 selected printed parts each have one solid.
model-audit.json

Parameter regeneration / PASS
Independent size trials; purchased dimensions checked and original geometry restored.
parameter-regeneration.json

Service paths / PASS
8 paths checked at poses no more than 1 mm apart. Prerequisites apply.
Final service-paths report; README

Driver shafts / PASS
14 approaches checked. Handles, hands and torque remain unqualified.
service-drivers.json

Selected walls and fasteners / PASS
20 sections and 14 screw/insert pairs checked. Global minimum wall is not verified.
selected-wall-fastener-checks.json

PCB allocation / PARTIAL
119 footprints / 115 FIT. Offline courtyard packing only; routing, thermal layout and real connectors remain open.
pcb-strip-assessment.json

Physical qualification / OPEN
Actual retention, global minimum wall, seals, gas response, wiring and prototype fit are not qualified.
README.md; MEASUREMENTS.md

Service intent: cover off and battery unplugged first. Pack and PCB carrier have separate rearward routes. Release external fittings before withdrawing the closed top cartridge. USB service follows pack removal; actual checked movements and prerequisites are recorded in README and the service report.

Concept deliverables only. Print-ready release follows actual part measurements, PCB layout, retention/seal checks and physical fit testing.